Lab Publications Retrieval Script: Report

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Purpose

The script `get_publications.py` automatically collects every publication by the lab's 10 members (5 permanent staff, 5 PhD candidates) since 1 January 2022. It returns one deduplicated, sortable table. This covers the first part of the task, retrieving publications; deduplication research and co-authorship analysis come next.

Data sources

The script combines three free bibliographic APIs, because no single one is complete. Google Scholar was left out because it has no public API.

Source	What it provides	Why it is used
OpenAlex	Author profiles with institutions, works with full author lists and affiliation countries	Main source; affiliations help pick the right person and detect Azerbaijani co-authors
Crossref	The DOI registry, searched by ORCID iD and by name	Catches DOI papers that OpenAlex misses
ORCID	Works members list on their own ORCID profile	Adds papers the members recorded themselves

dblp was tested but dropped, because its site blocks automated requests.

How it works

For each member, the script runs these steps in order:

1. **Find the author profile in OpenAlex.** It searches every known spelling of the name (e.g. "Ulviyya" and "Ulviya", "El Kadhi" and "Elkadhi").
    - A profile is accepted only if the surname matches and the first name (or initial) matches.
    - When several people share a name, the profile with a lab institution (UFAZ, ASOIU, INSA Rouen, LITIS, Strasbourg, York) is chosen. If none has one, name matches are kept but marked "UNVERIFIED" for review.
    - All matching profiles are kept, because OpenAlex sometimes splits one person into several.
2. **Fetch works from OpenAlex** for those profiles, published since 2022.
3. **Collect ORCID iDs** from the OpenAlex profiles, or search ORCID directly if none is linked.
4. **Query Crossref** by ORCID iD (exact) and by name (fuzzy). A name-based result is kept only with proof it is the right person: a matching ORCID, a lab affiliation, or another lab member as co-author.

5. **Fetch works from ORCID** profiles.
6. **Merge duplicates.** Records with the same DOI, or the same normalized title when a DOI is missing, become one row. The row lists every member and source involved.
7. **Apply the rule for Prof. Zanni-Merk's papers** (next section).
8. **Compare with the website list.** Each paper is marked as already on the lab website or not, by DOI or by title.

Requests retry with back-off when an API is busy or rate-limited, so a temporary error does not stop the run.

Rule for Prof. Zanni-Merk's papers

Prof. Zanni-Merk's papers are counted only if at least one co-author is Azerbaijani (from UFAZ or ASOIU). All other members' papers are counted regardless of co-authors. The check runs in order of reliability:

1. Another lab member is a co-author.
2. A co-author's affiliation is in Azerbaijan (country code AZ in OpenAlex, or "Azerbaijan", "Baku", "UFAZ", "ASOIU" in Crossref).
3. A co-author has a typical Azerbaijani surname ending (-ov, -ova, -yev, -zade, -li, ...). This is the weakest signal, so these rows are marked "please check".

The reason is written in the `az_coauthor` column. Papers that fail all three checks are not deleted; they go to a separate sheet, "Excluded (Cecilia)", for manual review.

Output files

The script writes three files next to itself:

File	Contents
`publications.xlsx`	Sheet "Publications" (newest first), sheet "Excluded (Dr.Cecilia)", sheet "Author matches"
`publications.csv`	Same data as the "Publications" sheet, as plain text
`author_matches.csv`	Which OpenAlex and ORCID profile was used for each member, with institutions, for verification

Columns in "Publications":

Column	Meaning
publication_date, year	When the paper was published
title, authors, venue, type	Bibliographic details (type: conference paper, article, preprint etc)
member	Lab members who authored it

Column	Meaning
doi, url	Identifier and link
source	Which API(s) found it
on_website	"yes" if already listed on the lab website
az_coauthor	For Prof. Zanni-Merk's papers: why it was kept

Results

The latest run found **45 unique publications** since 2022. 21 of them are already on the lab website and 24 are not, so these are candidates to add after a check.

- **By year:** 5 in 2022, 5 in 2023, 9 in 2024, 17 in 2025, 9 in 2026 so far.
- **By type:** mostly conference papers (26) and journal articles (8), plus preprints, book chapters, dissertations and one erratum.
- **Sources:** 33 of the 45 were confirmed by at least two APIs. Crossref and ORCID each added one paper that OpenAlex missed.
- **Prof. Zanni-Merk's papers:** 10 kept, each with a recorded reason (7 via another lab member, 2 via an Azerbaijani affiliation, 1 via surname). 34 were moved to the "Excluded (Cecilia)" sheet.
- **Author profiles:** a lab-affiliated OpenAlex profile was found for all 10 members, and an ORCID profile for 8 of them.

Website papers the script does not find

The website lists 34 papers. The ones the script misses fall into these groups:

- Papers by former members or co-authors who are not in the current member list (e.g. sign language papers, S. El Zant's paper).
- Russian- and Azerbaijani-language journal and conference papers, which the free APIs do not index.
- Papers marked "accepted, to be published" or pre-prints, which are not in the databases yet.
- Conference abstracts and proceedings without a DOI.

These can be covered with a short manual check. They are not errors in the matching logic.

Limitations and next steps

- **Coverage depends on the databases.** Papers missing from OpenAlex, Crossref and ORCID cannot be found automatically. Google Scholar would add some, but it has no API and was out of scope.
- **Duplicate detection is simple.** Two records merge only if they share a DOI or an identical normalized title. A preprint and its published version with a slightly different title stay as two rows.

- **Name matching is rule-based.** It handles spelling variants listed in the script but may miss unusual transliterations. Profiles marked "UNVERIFIED" should be checked by hand.
- **The surname rule is a heuristic.** Rows marked "surname (please check)" need a manual look as people from some countries such as Russia have similar surnames to Azerbaijani people sometimes.

The next parts of the task are researching better author deduplication and analysing co-authorship and research themes. The current name-matching and merge logic is a starting point for both.

How to run

1. Install the dependencies: `pip install requests pandas openpyxl`
2. Get a free OpenAlex API key at openalex.org/settings/api and paste it into `OPENALEX_API_KEY` at the top of the script.
3. Keep `website_publications.txt` in the same folder to get the `on_website` column.
4. Run `python get_publications.py`. A full run takes a few minutes.

To add or update a member, edit the `MEMBERS` list at the top of the script. To change the start year, edit `FROM_YEAR`.